

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA1008	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
Student Name:	B. Pushparaj	Reg. No.:	192421136
Date:	13.08.2026	Duration:	60 minutes

Code of Conduct

I **B. Pushparaj (192421136)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____



Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. Analyze System Requirements

- Study the given problem statement and identify the system objectives.
- Analyze the functional and non-functional requirements that influence the software architecture.

- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.
- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
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System Requirement Analysis	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.
Component Interaction and Quality Attribute Analysis	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

	well-organized, professional report.			
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Criteria	Mark
System Requirement Analysis	/5
Selection of Software Architecture	/5
Architecture Diagram and Component Design	/5
Component Interaction and Quality Attribute Analysis	/5
Architectural Justification and Technical Presentation	/5
TOTAL	/5

1. Introduction and Problem Statement

Students with visual impairments often face difficulties while using conventional online learning platforms because many websites and educational resources are not designed for accessibility.

The proposed **Accessible Learning Platform** is a web-based educational system designed to provide equal learning opportunities for students with visual impairments. The platform supports **screen readers, voice navigation, audio learning resources, keyboard navigation, accessible quizzes, course materials, progress tracking, and teacher interaction.**

The system allows students to access learning materials using both visual and audio interfaces. Teachers can upload accessible learning resources, create quizzes, monitor student progress, and provide feedback.

The main goal is to make online education **accessible, easy to navigate, and usable for students with different accessibility requirements.**

2. System Requirements Analysis

2.1 System Objectives

1. Provide an accessible online learning environment for students with visual impairments.
2. Support screen-reader-compatible web pages.
3. Provide voice-based navigation for important platform functions.
4. Provide audio versions of learning materials.
5. Allow students to attend accessible online quizzes and assessments.
6. Allow teachers to upload courses, notes, audio resources, and quizzes.
7. Track student learning progress and assessment results.
8. Provide administrators with tools to manage users and learning content.
9. Maintain student data securely.
10. Provide a simple and user-friendly interface.

2.2 Functional Requirements

1. The system shall allow students and teachers to register and log in.
2. The system shall provide role-based access for students, teachers, and administrators.
3. Students shall be able to browse and access available courses.
4. Students shall be able to listen to audio learning materials.

5. The system shall support keyboard navigation.
6. The system shall support screen-reader-compatible content.
7. Students shall be able to use voice commands for navigation.
8. Students shall be able to attend accessible quizzes.
9. The system shall automatically calculate quiz scores.
10. Students shall be able to view their learning progress.
11. Teachers shall be able to upload learning materials.
12. Teachers shall be able to create and manage quizzes.
13. Teachers shall be able to view student performance.
14. Administrators shall be able to manage users and courses.
15. The system shall generate progress reports.

2.3 Non-Functional Requirements

Accessibility

The platform should be usable with screen readers, keyboard navigation, audio controls, and voice navigation.

Usability

The interface should be simple, consistent, and easy to understand.

Security

Student accounts, passwords, quiz results, and learning records must be protected.

Performance

Pages, audio resources, quizzes, and course content should load quickly.

Reliability

The platform should remain available during normal learning activities and prevent loss of student progress.

Scalability

The system should support an increasing number of students, teachers, courses, and learning resources.

Maintainability

New accessibility features and learning modules should be easy to add.

Availability

The platform should be accessible whenever students need to study.

2.4 Key Quality Attributes

Quality Attribute	Importance
Accessibility	Allows students with visual impairments to use the platform independently
Usability	Makes navigation simple and understandable
Security	Protects student and teacher information
Performance	Provides fast access to learning materials
Reliability	Prevents loss of learning progress and quiz results
Scalability	Supports more users and courses in the future
Maintainability	Makes future modifications easier
Availability	Allows students to access learning resources whenever required

3. Selection of Suitable Software Architecture

3.1 Selected Architecture

Layered Architecture with Client-Server Model

The proposed Accessible Learning Platform uses a **Layered Architecture combined with a Client-Server approach**.

The major layers are:

1. Presentation Layer
2. Application Layer
3. Accessibility Services Layer
4. Learning Management Layer
5. Data Layer

This structure separates the user interface, business logic, accessibility functions, learning functions, and database.

3.2 Why This Architecture is Suitable

1. Easy Accessibility Integration

Accessibility services such as screen-reader support, voice navigation, keyboard navigation, and text-to-speech can be maintained separately.

2. Easy Maintenance

Changes to the user interface do not require major changes to the database.

3. Security

Authentication and authorization can be handled at the application layer.

4. Scalability

More students, courses, and learning resources can be added without redesigning the entire system.

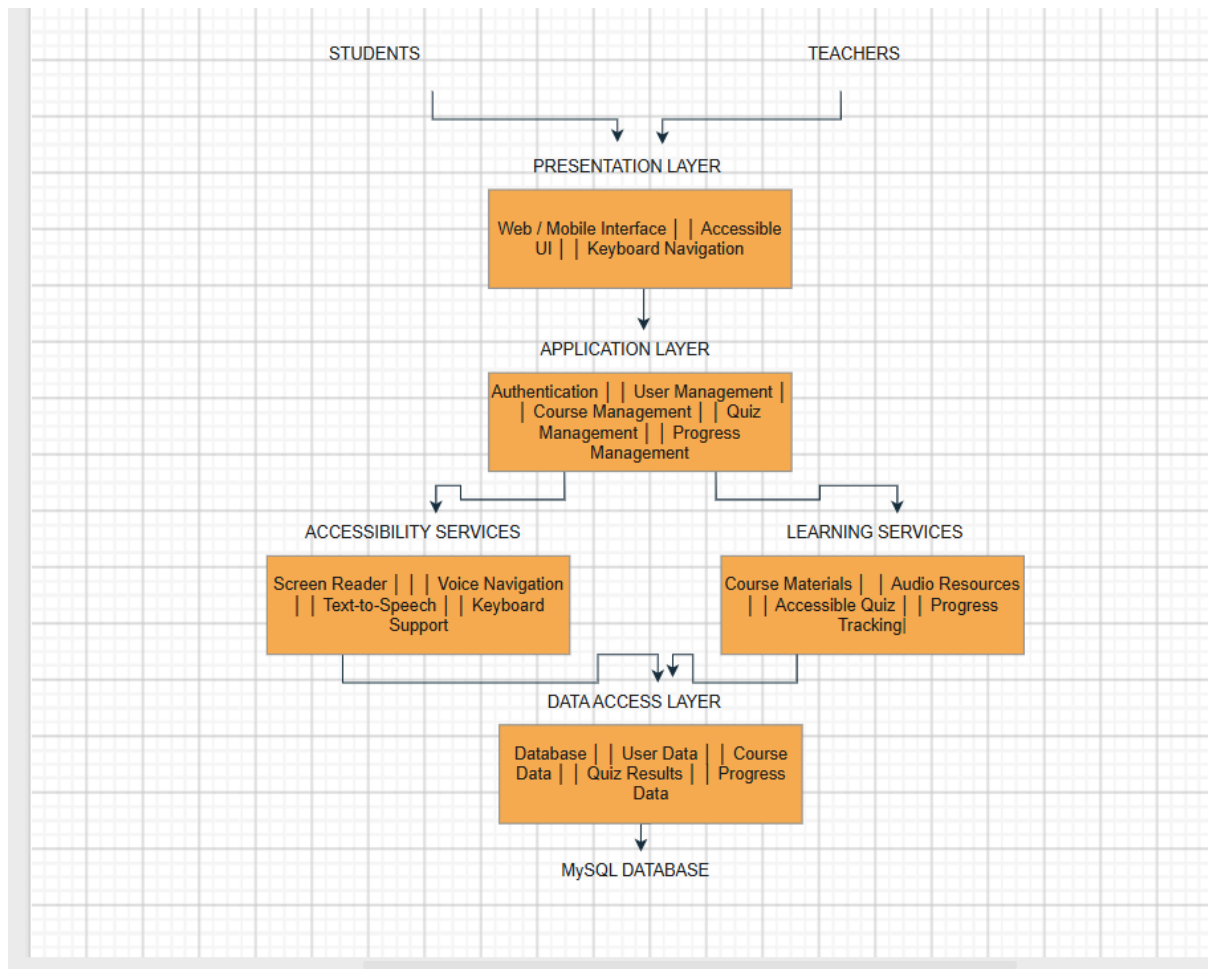
5. Simple Development

The architecture is suitable for an academic project because it is easier to develop, test, and understand than a complete microservices architecture.

6. Future Expansion

Additional services such as mobile applications, advanced accessibility tools, and external educational systems can be integrated later.

4. Software Architecture Diagram



5. Components and Their Interactions

5.1 Presentation Layer

The Presentation Layer is the part directly used by students, teachers, and administrators.

Components

- Login/Register Page
- Student Dashboard
- Teacher Dashboard
- Course Page
- Quiz Page
- Progress Page
- Admin Dashboard

The interface uses proper headings, labels, buttons, keyboard navigation, and alternative text so that screen readers can understand the content.

5.2 Accessibility Services Layer

This is the most important layer for the proposed system.

Screen Reader Support

The web pages are designed using semantic HTML elements so screen readers can identify headings, buttons, links, forms, and content.

Voice Navigation

Students can use commands such as:

"Open courses"

"Start quiz"

"Go to dashboard"

"Open progress"

"Next question"

Text-to-Speech

Text-based learning material can be converted into spoken audio.

Keyboard Navigation

Students can navigate through the platform using:

Tab → Move between controls

Enter → Select

Arrow Keys → Navigate options

Esc → Close dialogs

5.3 Learning Management Layer

This layer handles the main educational functions.

Course Management

Teachers can:

- Create courses
- Upload notes
- Upload audio materials
- Update course content
- Delete outdated materials

Quiz Management

Teachers can:

- Create questions
- Add multiple-choice options
- Set correct answers
- Set quiz duration
- View student scores

Progress Management

The system records:

- Courses completed
- Quiz scores
- Attempts
- Learning progress
- Completed lessons

5.4 Application Layer

The Application Layer contains the main business logic.

Authentication Module

Handles:

- Login
- Registration
- Logout
- Password validation
- Role-based access

User Management

Manages:

- Student accounts
- Teacher accounts
- Administrator accounts

Course Management

Controls course creation, updating, viewing, and deletion.

Assessment Management

Processes quiz attempts and calculates scores.

Progress Management

Updates student progress based on completed lessons and quiz results.

5.5 Data Layer

The Data Layer stores all system information.

Main Database Tables

Users

Courses

Lessons

Audio_Materials

Quizzes

Questions

Quiz_Results

Progress

Accessibility_Settings

The database can be implemented using **MySQL**.

5.6 Overall Workflow

The overall workflow is:

Student



Login



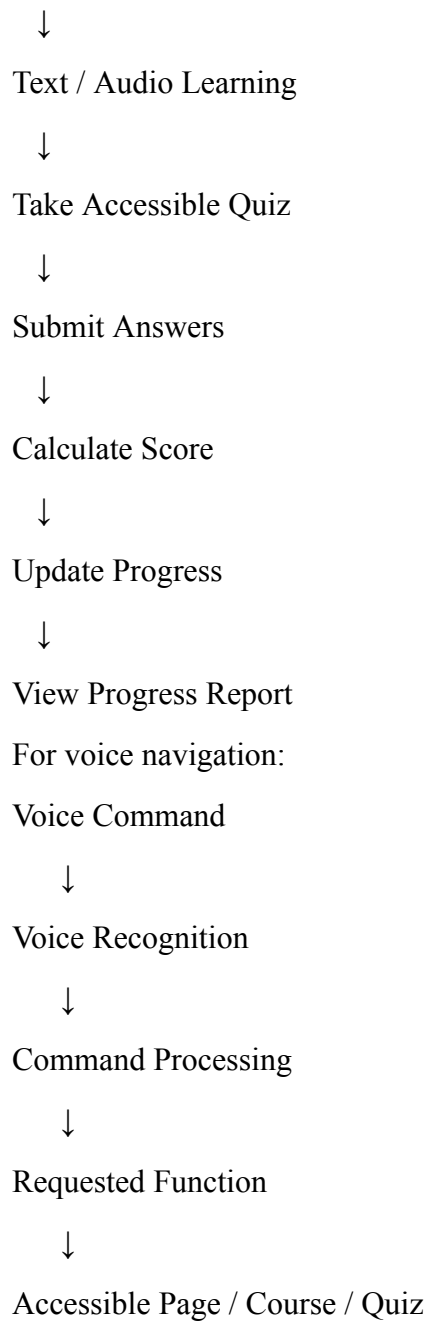
Student Dashboard



Select Course



Access Learning Material



6. Discussion of Quality Attributes

6.1 Accessibility

Accessibility is the primary quality attribute.

The system supports:

- Screen readers
- Voice navigation
- Text-to-speech

- Keyboard navigation
- Accessible forms
- Proper heading structure
- Alternative text
- Audio learning materials

6.2 Security

The system protects user information through:

- Password authentication
- Role-based access
- Secure sessions
- Input validation
- Database access control

Students cannot access teacher or administrator functions.

6.3 Performance

Performance can be improved through:

- Optimized database queries
- Compressed audio files
- Efficient page loading
- Browser caching
- Lightweight web pages

6.4 Reliability

The system should save:

- Quiz results
- Student progress
- Course completion status
- User information

This prevents students from losing their learning progress.

6.5 Scalability

The platform can be expanded to support:

- More students
- More teachers
- More courses
- More audio resources
- Mobile applications
- Multiple educational institutions

6.6 Maintainability

The layered architecture makes maintenance easier because each layer has a specific responsibility.

For example:

Change UI



Presentation Layer

Change Voice Feature



Accessibility Layer

Change Quiz Logic



Application/Learning Layer

Change Database



Data Layer

6.7 Availability

The platform should be available throughout the day so students can access their courses and assessments whenever required.

Regular database backup can be used to prevent data loss.

7. Justification of Architectural Decisions

7.1 Rationale

The Layered Client-Server Architecture is suitable because the Accessible Learning Platform contains several clearly separated responsibilities.

The Presentation Layer handles the user interface.

The Accessibility Layer handles screen readers, voice navigation, text-to-speech, and keyboard support.

The Learning Layer handles courses, lessons, quizzes, and progress.

The Data Layer stores user and learning information.

This separation makes the system easier to understand, develop, test, and maintain.

7.2 Trade-offs Considered

Layered Architecture vs Microservices

A microservices architecture could provide independent scaling, but it would introduce unnecessary complexity for this academic project.

Therefore, layered architecture is selected because it is:

- Simpler
- Easier to develop
- Easier to test
- Easier to maintain
- Suitable for the expected project size

Web Application vs Mobile Application

A web application is selected as the primary platform because students can access it through computers and mobile browsers without installing an application.

A dedicated mobile application can be added later.

Local Text-to-Speech vs External Service

Browser-based text-to-speech can be used initially to keep the project simple. External text-to-speech services can be integrated later if better voice quality or additional languages are required.

7.3 Future Enhancements

The system can be enhanced with:

1. Mobile application support.
2. More regional language audio support.
3. Offline learning materials.
4. Braille-compatible learning resources.
5. Voice-based quiz answering.
6. Teacher-student voice interaction.
7. Integration with existing Learning Management Systems.
8. Accessibility compliance testing.
9. Downloadable accessible study materials.
10. Personalized accessibility settings such as font size, contrast, speech rate, and audio controls.

8. Conclusion

The proposed **Accessible Learning Platform** provides an inclusive digital learning environment for students with visual impairments.

The system uses a **Layered Client-Server Architecture** consisting of Presentation, Application, Accessibility Services, Learning Services, and Data layers.

The architecture directly supports the major requirements of the project, including **screen readers, voice navigation, text-to-speech, audio learning materials, keyboard navigation, accessible assessments, progress tracking, and secure data management**.

The selected architecture provides good **accessibility, security, performance, reliability, scalability, maintainability, and availability** while remaining simple enough for implementation as a student software engineering project.

The structure also follows the architecture-assignment requirements in the reference document, including system requirements, architecture selection, diagram, component interaction, quality attributes, and architectural justification.